



D.Y. Patil Academic Education Excellence Federation's

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Polytechnic | Engineering | MCA

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## DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

TOPIC: Microprocessor and Microcontroller(8051)

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## Microprocessor and Microcontroller (8051)

### 1. Introduction to Microprocessor and Microcontroller

**Microprocessor:** A microprocessor is a programmable electronic chip that mainly contains the CPU. It performs arithmetic, logical and control operations and normally requires external memory and I/O devices.

**Microcontroller:** A microcontroller is a complete small computer on a single chip. It contains a CPU, memory, I/O ports and peripherals such as timers and serial communication interfaces.

**Key point:** Microprocessor = mainly CPU + external components; Microcontroller memory + I/O +peripherals.

### 2. What Is a Microprocessor?

A microprocessor executes instructions stored in memory and controls the operation of a digital system.

Examples include 8085 and 8086.

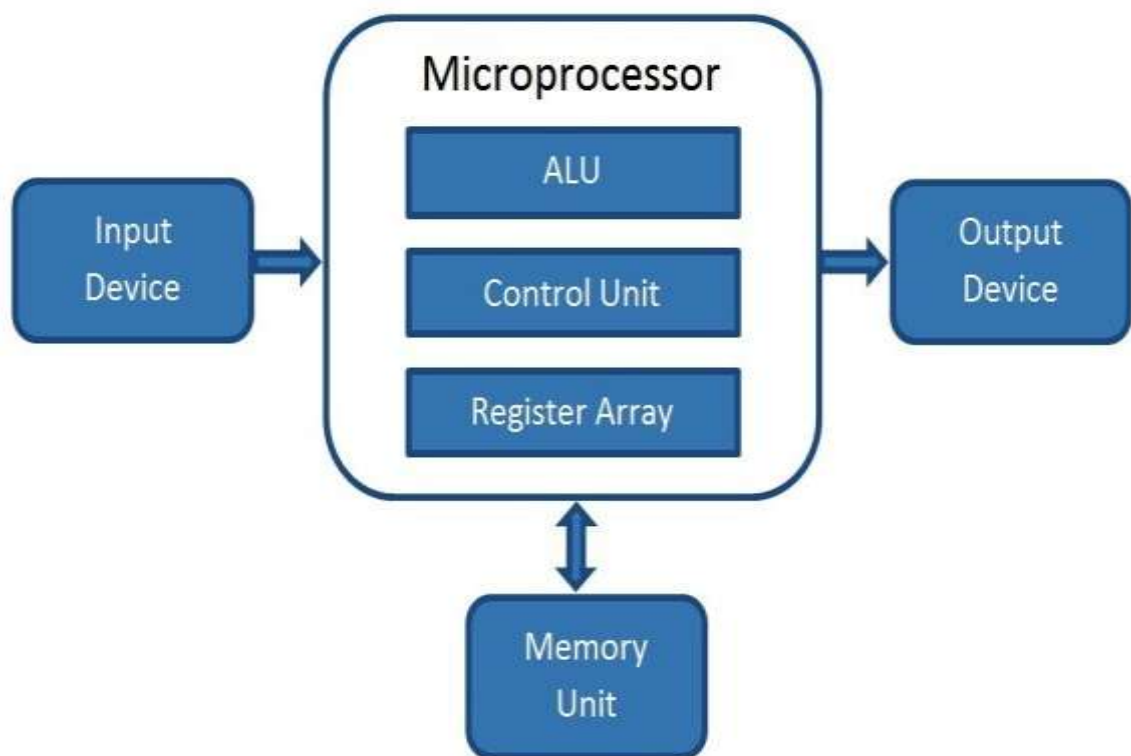
Main components: ALU, Control Unit, Registers and buses.

Applications: computers, general-purpose processing systems, and systems requiring high processing flexibility.

### 3. Advantages of Microprocessor

- High processing speed
- Performs complex calculations
- Flexible and programmable
- Supports large memory
- Suitable for multitasking
- Easy to upgrade
- Used in computers and advanced systems
- Reliable and compact

### 4. Block Diagram of Microprocessor



### 5. What Is a Microcontroller?

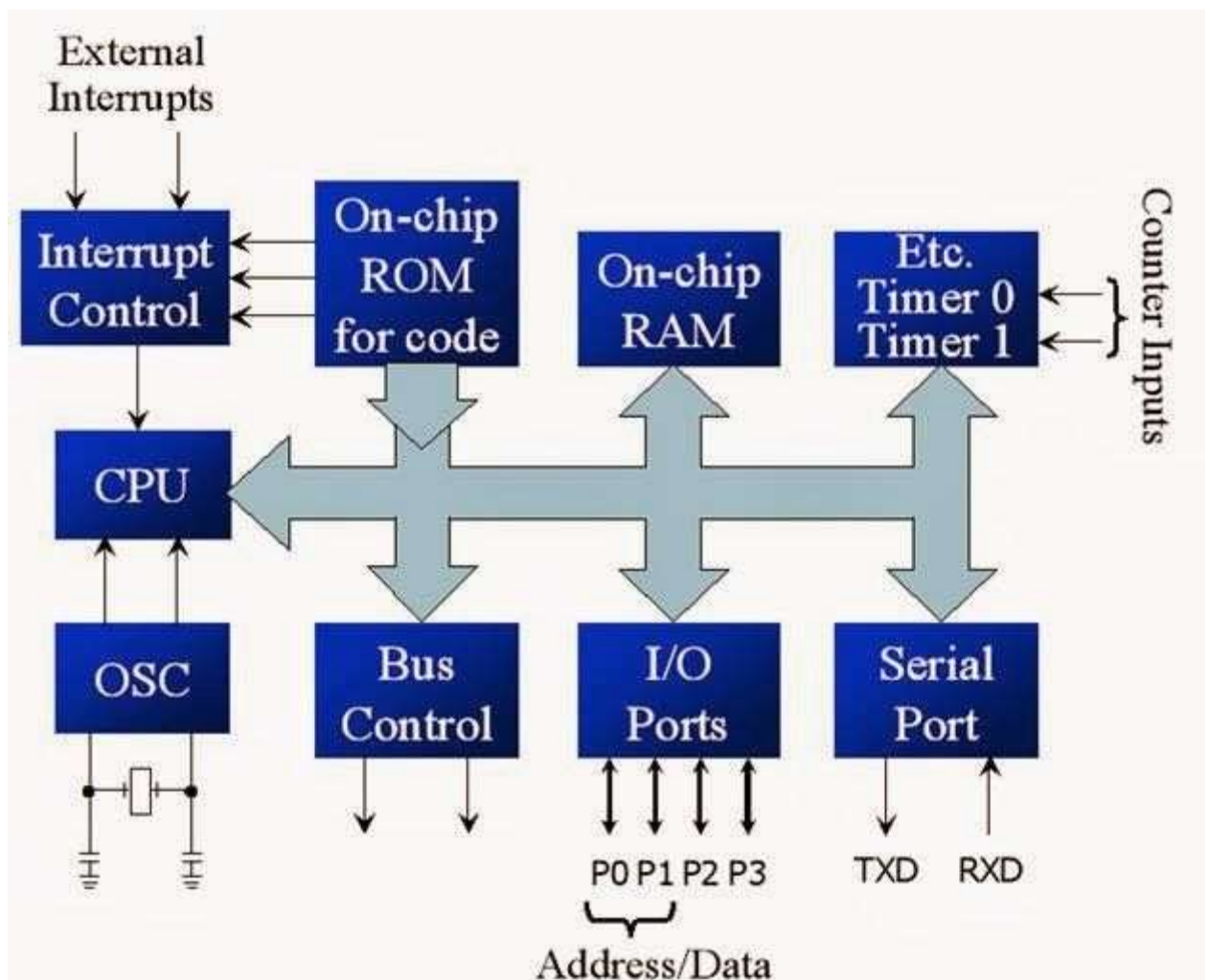
A microcontroller integrates processing, memory and control peripherals into one IC. It is mainly used in embedded systems where a dedicated task must be controlled. Main components: CPU, RAM, ROM/Flash, I/O ports, timers/counters, interrupts and serial communication.

Applications: automation, appliances, traffic control, sensor systems, robotics and IoT devices.

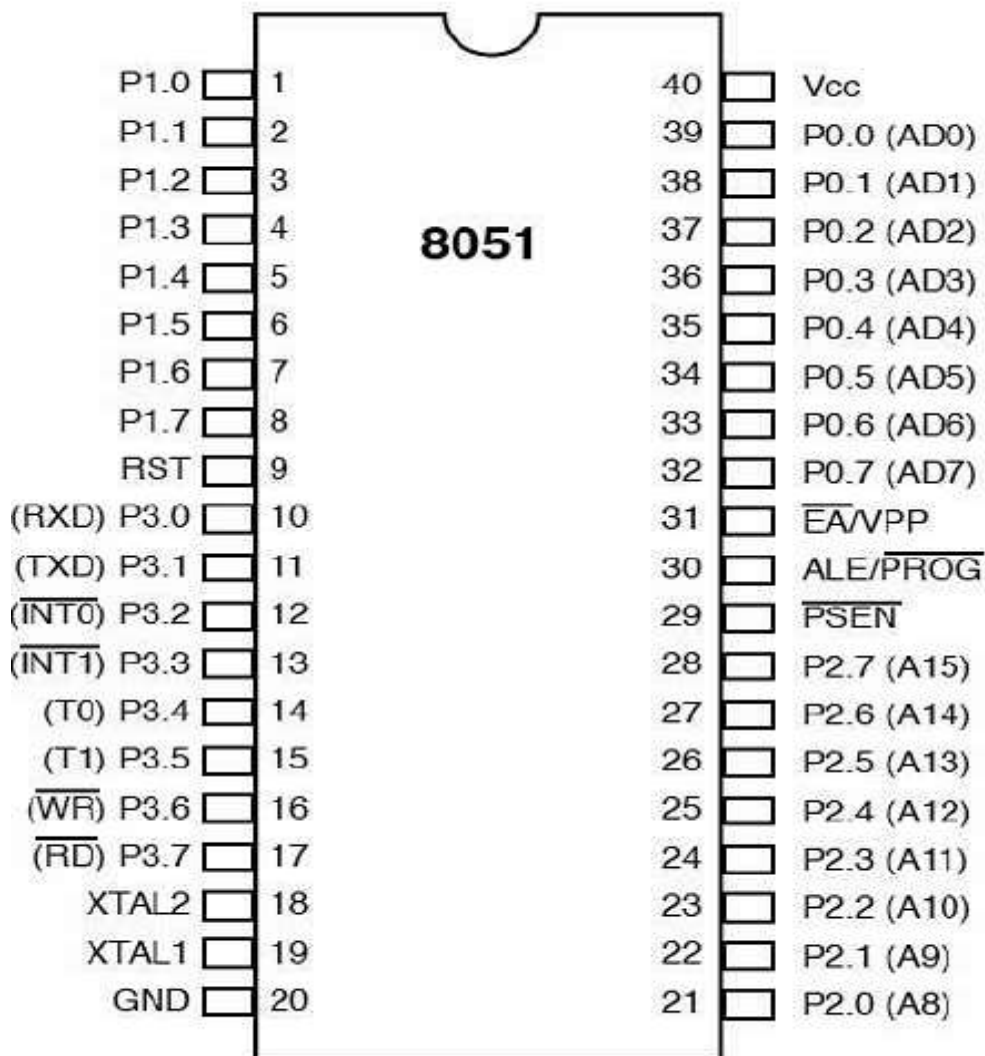
## 6. Advantages of Microcontroller

- Low cost
- Low power consumption
- Compact size
- Built-in memory and peripherals
- Requires fewer external components
- Easy to program
- Fast response for control applications
- Reliable and suitable for embedded systems

## 7. Block Diagram of Microcontroller



## 8. Pin Diagram of 8051



- 8051 has 40 pins.
- Port 0: I/O and address/data bus.
- Port 1: General-purpose I/O.
- Port 2: I/O and address bus.
- Port 3: I/O with special functions.
- VCC & GND: Power supply.
- RST: Resets the microcontroller.
- XTAL1 & XTAL2: Provide clock signal.

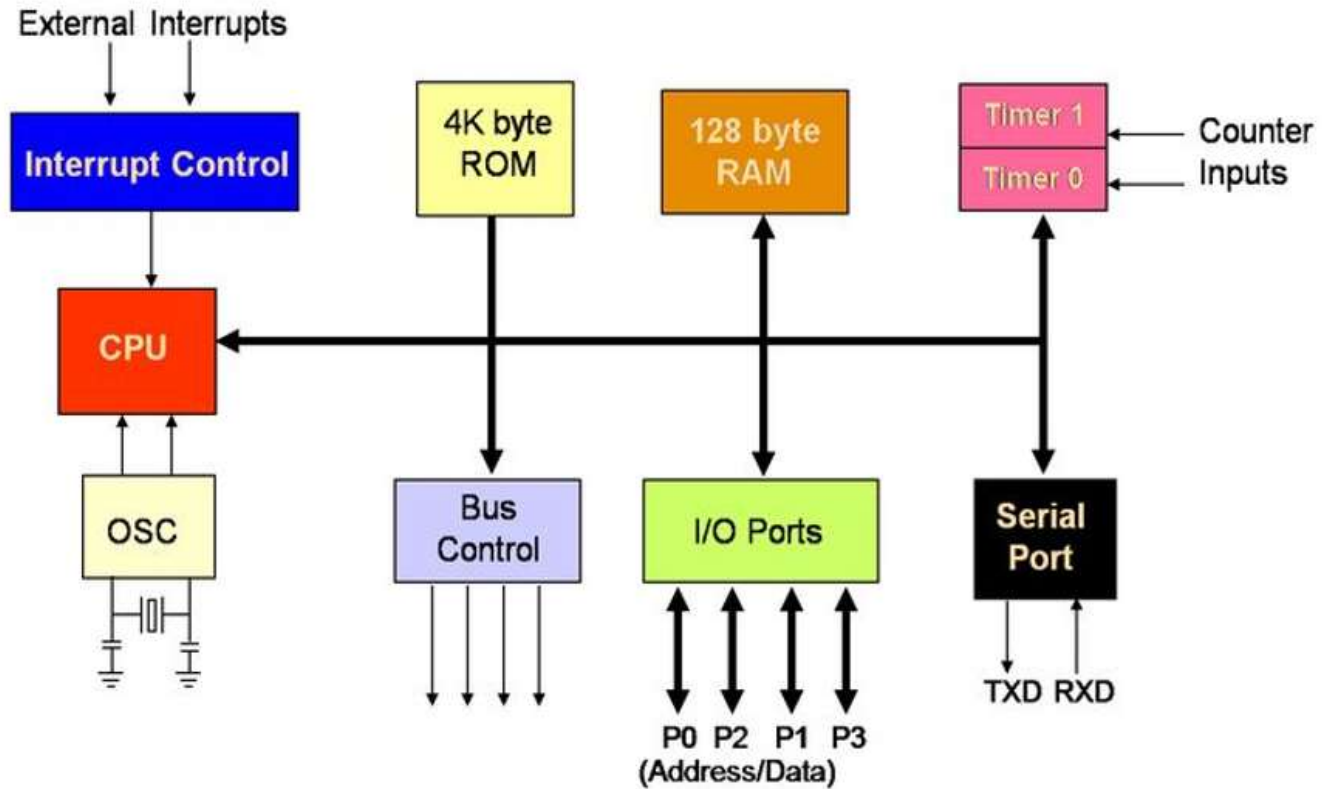
- EA, ALE & PSEN: External memory control.
- Instruction Set
  - An instruction set is a collection of commands understood by a processor.
  - Instructions perform operations such as addition, subtraction and data transfer.
  - Microprocessors often support complex instruction sets.
  - Microcontrollers use instruction sets optimized for control applications
- Registers
  - Registers are small, high-speed storage locations inside the processor.
  - They temporarily store data, addresses and instructions.
  - Common registers include the accumulator, program counter and status register.
  - Registers help improve processing speed.
- Input and Output Ports
  - I/O ports allow the controller to communicate with external devices.
  - Input devices include sensors, switches and keyboards.
  - Output devices include LEDs, motors, displays and relays.
  - Ports can often be configured as input or output.

## 9. Working 8051

- The clock provides timing.
- CPU fetches instructions from memory.
- Instructions are decoded and executed.
- Data is processed and the result is given through output ports.

Flow: Input → Processing → Output

## 10.8051 Block Diagram



## 11. Main Features of Classic 8051

- 8-bit CPU+
- 4 KB on-chip program ROM
- 128 bytes internal RAM
- 32 programmable I/O lines
- Two 16-bit timers/counters
- Full-duplex serial port
- Interrupt system
- Oscillator/clock circuitry

## 12. Microprocessor vs Microcontroller

### Microprocessor vs Microcontroller

|                                                                                                          |                                                                                                                                     |                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Memory                                                                                                   | Needs external memory & storage, flexible                                                                                           | On-chip memory (RAM, ROM, EEPROM)                                                                                                        |
| Peripheral<br>          | Need additional part via external bus, ideal for complex, powerful applications.                                                    | Integrated peripheral SPI, UART, RTI, ADC for direct HW control                                                                          |
| Clock Speed<br>        | Fast GHz for heavy duty tasks<br>                 | KHz - MHz, efficient for specific tasks.<br>         |
| Power Consumption<br> | High power consumption, no low power option.<br> | Low power consumption with power saving options<br> |
| Connectivity                                                                                             | Handles high-speed transfer                                                                                                         | supports low to moderate speed communication I2C, SPI, UART                                                                              |
| Operating System<br>  | Need full-fledge OS (Linux, Windows)                                                                                                | OS is optional (no-OS or light weight RTOS)                                                                                              |
| Usecase                                                                                                  | For generic computing, high complexity capacity system                                                                              | For compact systems, battery powered, or logic processing devices                                                                        |
| Cost                                                                                                     | Higher cost — suited for performance-driven applications                                                                            | Lower cost — economical for mass production.                                                                                             |

## **13. Applications of 8051**

- Traffic light control
- Digital clock and counters
- Temperature monitoring
- Motor control
- Security systems
- LCD and keypad interfacing
- Industrial automation
- Basic IoT and sensor-control projects

## **14. Advantages**

- Simple and easy to learn
- Compact embedded solution
- Built-in timers and serial communication
- Multiple programmable I/O pins
- Suitable for basic control applications

## **15. Future Trends**

- Artificial intelligence at the edge
- Internet of Things
- Smart sensors
- Autonomous vehicles
- Wearable electronics
- Energy-efficient processors

## **16. Limitations**

- Limited Memory – Classic 8051 has only 4 KB ROM and 128 bytes RAM, which is small for complex applications.
- 8-bit Processor – It processes 8-bit data, so it is slower for large or complex calculations compared with modern 16-bit and 32-bit microcontrollers.
- Limited Processing Speed – The original 8051 operates at relatively low clock speeds compared with modern microcontrollers.
- Limited Built-in Peripherals – The classic 8051 has fewer built-in features such as ADC, DAC, and advanced communication interfaces.
- External Components Required – Some applications need external memory, ADC, drivers, or other hardware, increasing circuit complexity.

## 17. Conclusion

The 8051 is a popular 8-bit microcontroller used in embedded systems. It combines CPU, memory, I/O ports, timers, interrupts, and serial communication on a single chip. It is simple, low-cost, compact, and easy to program. It is widely used in traffic control, digital clocks, security systems, motor control, home automation, and industrial applications. Although it has limited memory and processing power, it is very useful for basic control and automation applications. Learning 8051 provides a strong foundation for understanding modern microcontrollers and embedded systems.

## 18. References

- NPTEL – Introduction to Microcontrollers & 8051 Covers 8051 architecture, ports, timers, interrupts, serial communication, and instruction set
- [https://archive.nptel.ac.in/content/storage2/courses/117104072/micro/lecture5/lec5\\_1.htm?utm\\_source=chatgpt.com](https://archive.nptel.ac.in/content/storage2/courses/117104072/micro/lecture5/lec5_1.htm?utm_source=chatgpt.com)